

CS & IT ENGINEERING



Operating System

Virtual Memory

Lecture – 5

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Recap of Previous Lecture



Topic

LIFO, LFU, MFU, MRU Policies

Topic

Page Replacement Algorithms

Topics to be Covered



Topic

Frame Allocation

Topic

Thrashing

Topic

Multilevel Paging



Topic : Frame Allocation





Topic : Frame Allocation

- How many frames do we allocate per process?



Topic : Frame Allocation

How many frames do we allocate per process?

- If it is a single-user, single-tasking system, it's simple – all the frames belong to the user's process



Topic : Frame Allocation

2 Questions

- What is the minimum number of frames that a process needs?
- Is page replacement global or local?



Topic : Minimum Number of Frames

Every process must have enough pages to complete an instruction.



Topic : Frame Allocation

1. Equal Allocation $\Rightarrow$ allocate equal no. of frames to each process
2. Proportional Allocation $\Rightarrow$ allocate frames, proportional to process's size

ex:-

no. of frames = 6

2 processes

P1 $\Rightarrow$ 8 pages

P2 $\Rightarrow$ 4 pages

equal allocation

P1 $\Rightarrow$ 3 frames

P2 $\Rightarrow$ 3 frames

Proportional allocation

$$P1 \Rightarrow \frac{8}{(8+4)} * 6 = 4$$

$$P2 \Rightarrow \frac{4}{(8+4)} * 6 = 2$$



Topic : Local Vs Global Allocation

↓
at the time
of page fault for
a process P , the
new page can replace
only existing pages
of P only.

↓
a new incoming page can replace
a page of other process also



Topic : Local Allocation

1. Local replacement requires that the page being replaced be in a frame belonging to the same process
2. The number of frames belonging to the process will not change
3. This allows processes to control their own page fault rate

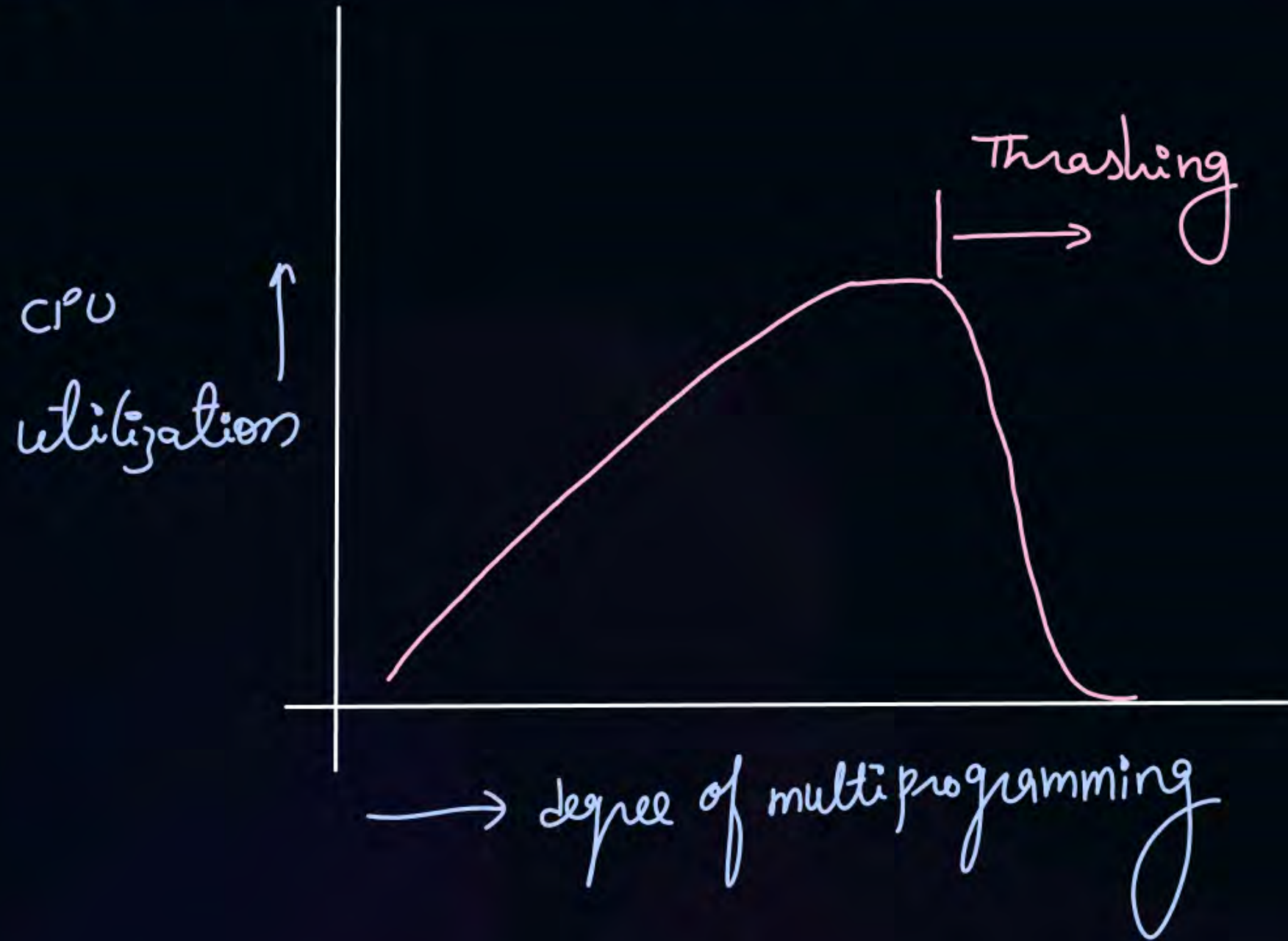


Topic : Global Allocation

1. The process can replace a page from a set that includes all the frames allocated to user processes
2. High-priority processes can increase their allocation at the expense of lower-priority processes
3. Global allocation makes for more efficient use of frames and their better throughput



Topic : Thrashing





Topic : Locality Model

1. Working Set Model
2. Page Fault Frequency



Topic : Working Set



↓
get the no. of pages needed to run a process without higher
page fault rate and provide those many frames to
the process.



Topic : Page Fault Frequency





Topic : Let's Take a Simple Example

Process Size = 32B $\leftarrow$ logical address space

Page size = 4B

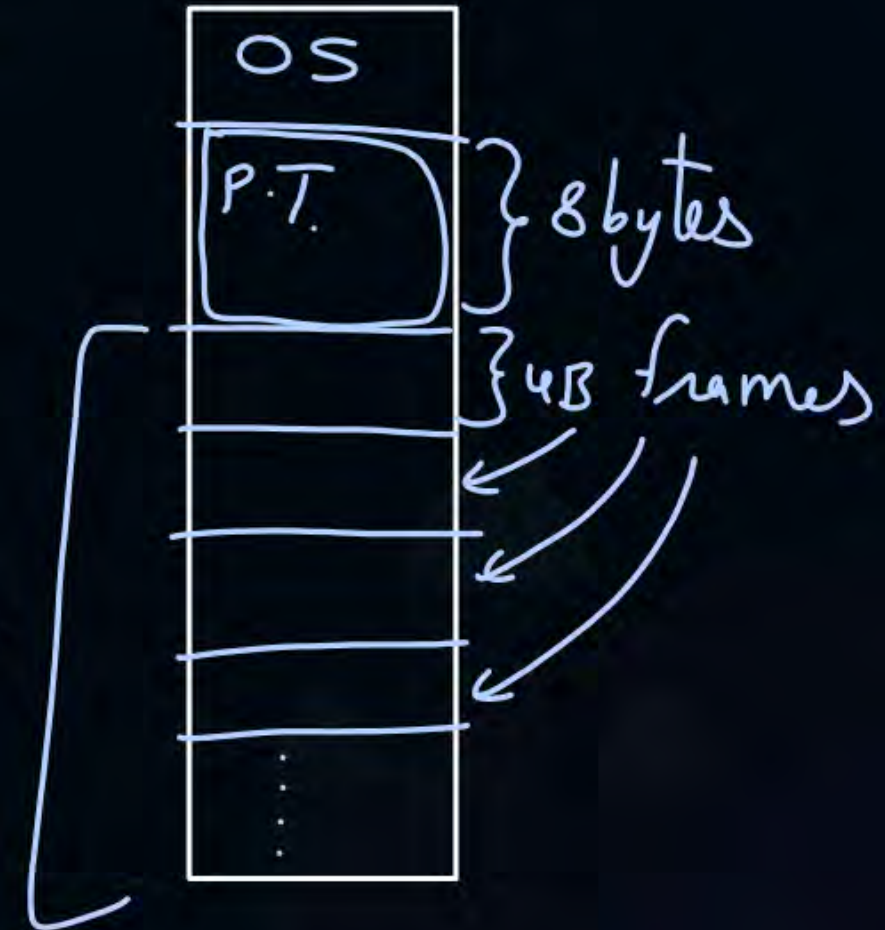
Page table entry size = 1B

Page Table Size = ?

$$\text{no. of pages} = \frac{32B}{4B} = 8$$

$$\begin{aligned} \text{P.T. size} &= 8 * 1B \\ &= 8 \text{ bytes} \end{aligned}$$

frames



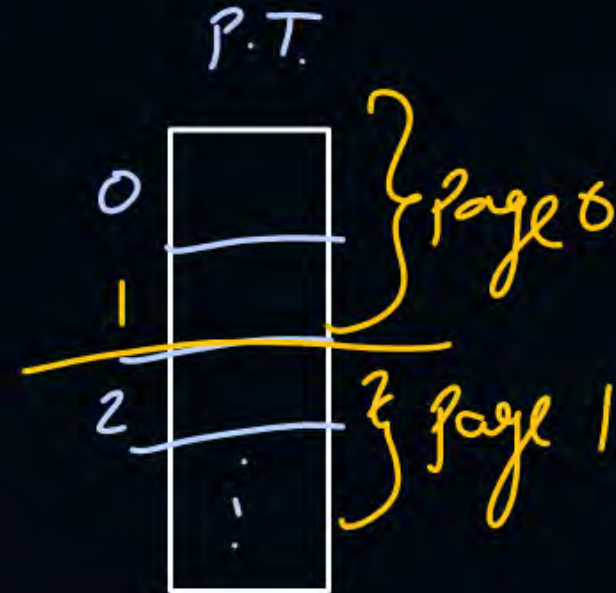


Topic : Now What?

Can you keep entire page table in single page? $\Rightarrow$ no $\Rightarrow$ 8 byte P.T. should be stored on 2 frames of 4 Bytes each

If not then how to access the specific entry?

P.T. for P.T.





Topic : Page Table in Memory

Process
Page 000
Page 001
Page 010
Page 011
Page 100
Page 101
Page 110
Page 111

	Page Table
000	0100
001	1000
010	1101
011	1010
100	0101
101	0111
110	0001
111	1100

Page 0 of P.T.

Page 1 of P.T.

outer P.T.

0010
1111

(Inner P.T.)

Page 0 of P.T.

00	0100
01	1000
10	1101
11	1010

Page 1 of P.T.

00	0101
01	0111
10	0001
11	1100

main mem.

0000	
0001	Page 110
0010	Page 0 of PT
0011	
0100	Page 000
0101	Page 100
0110	
0111	Page 101
1000	Page 001
1001	
1010	Page 011
1011	
1100	Page 111
1101	Page 010
1110	
1111	Page 1 of PT



Topic : Multilevel Paging

if CPU generates LA

P
010
d

0	10	
P1	P2	

↓
goto page 0 of P.T. & get P.T. entry no. 10 in that

LA

101	d
1.	01



Topic : Multilevel Paging

logical address

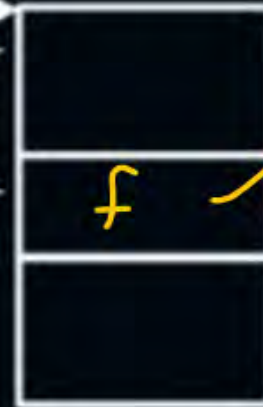


P_1 {



outer -page table

P_2 {

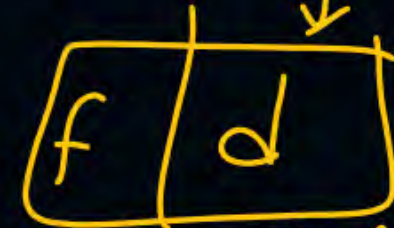


outer -page table

d {



desired page



physical address



Topic : Multilevel Paging



ex:-

Process size = 256 B, LA = 8 bits

Page size = 16 Bytes, d = 4 bits

one P.T. entry = 2 bytes

no. of P.T. entries per page

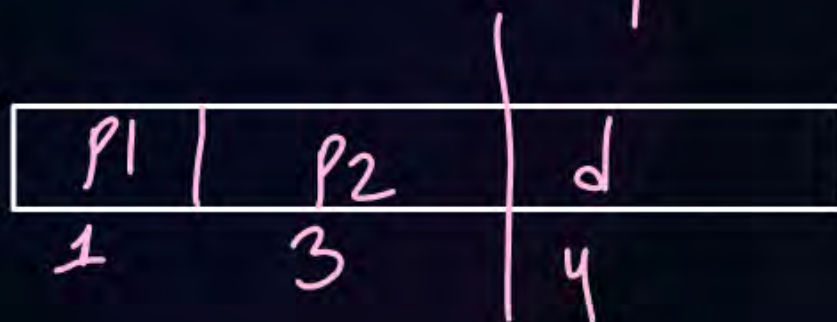
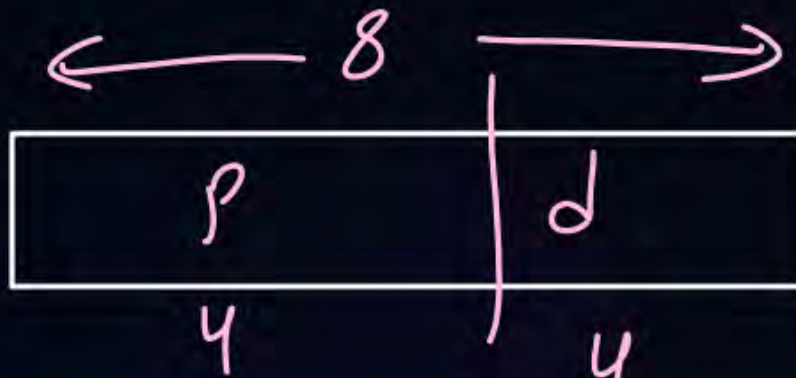
$$= \frac{\text{Page size}}{\text{P.T. entry size}}$$

$$= \frac{16 \text{ B}}{2 \text{ B}}$$

$$= 8 = 2^3$$

↓

no. of bits needed to search
in P.T. on one page = 3 bits



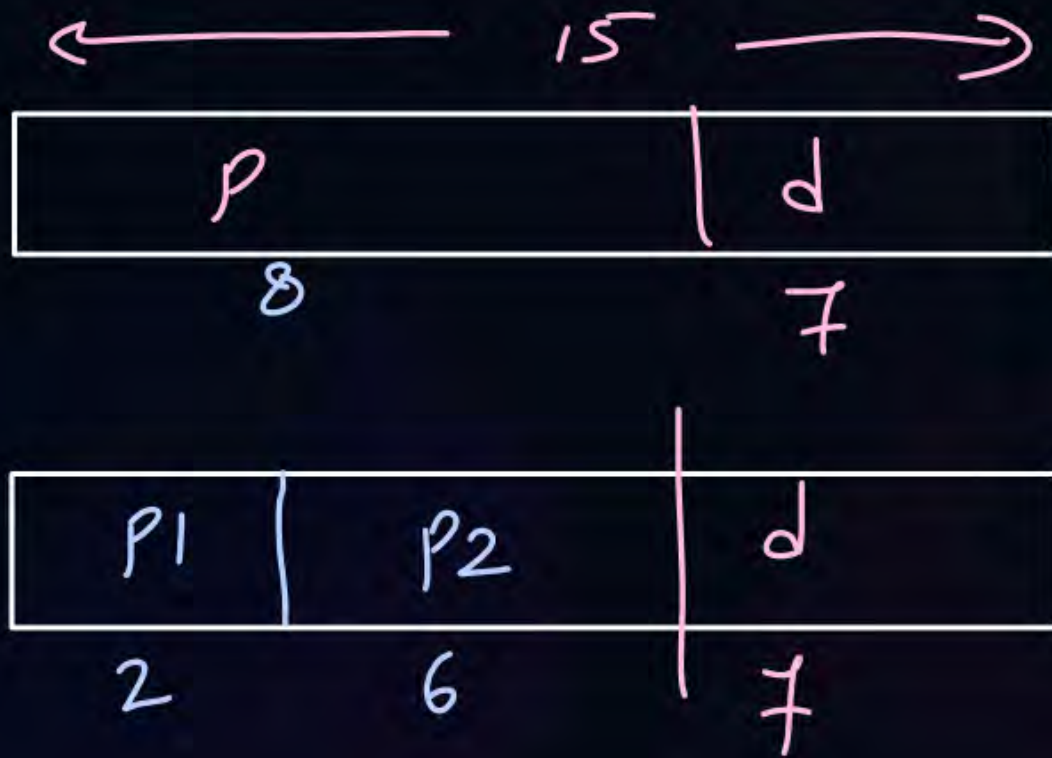


Topic : Question

VA = 15 bits
Page size = 128 bytes = 2^7 bytes
Page table entry size = 2 bytes

no. of P.T. entries per page
$$= \frac{128 \text{ B}}{2 \text{ B}} = 64 = 2^6$$

Number of levels in multilevel page 2?



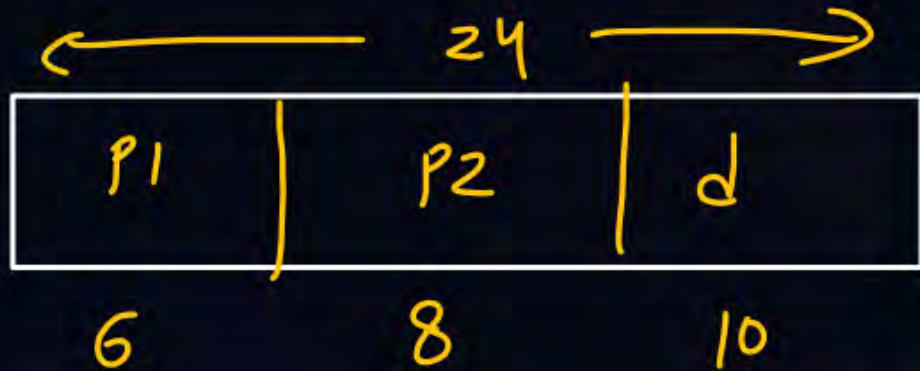


Topic : Question

VA = 24 bits
Page size = 1Kbytes = 2^{10} B
Page table entry size = 4 bytes

no. of P.T. entries
per page = $\frac{2^{10} \text{ B}}{4 \text{ B}}$
= 2^8

Number of levels in multilevel page 2?



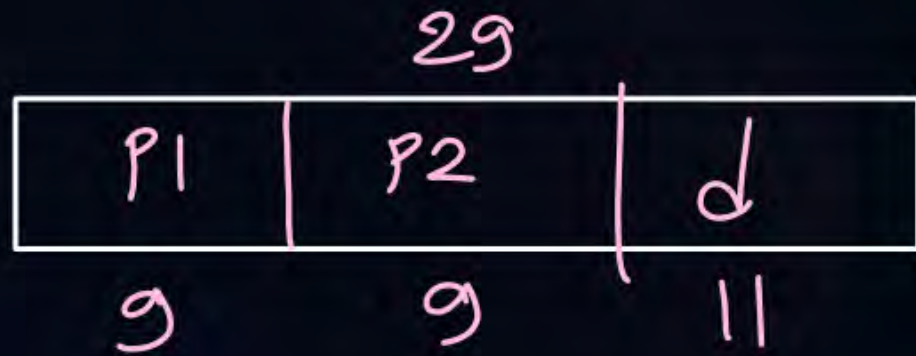


Topic : Question

VA = 29 bits
Page size = 2Kbytes = 2^{11}
Page table entry size = 4 bytes

$$\text{no. of P.T. entries per page} = \frac{2^{11} \text{ B}}{4 \text{ B}} = 2^9$$

Number of levels in multilevel page 2?



outer P.T. fits exactly in single page



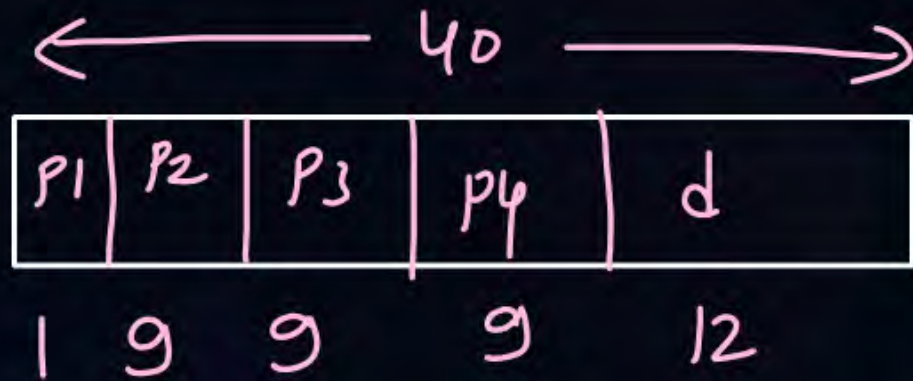
Topic : Question

VA = 40 bits

Page size = 4Kbytes = 2^{12}

Page table entry size = 8 bytes

Number of levels in multilevel page 4 ?



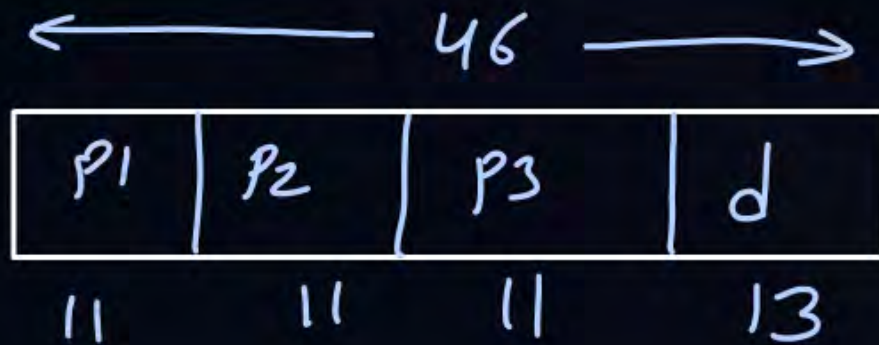
no. of P.T. entries

$$\text{per page} = \frac{2^{12} B}{8 B} = 2^9$$



Topic : Question

#Q. Consider a virtual memory system with physical memory of 8GB, a page size of 8KB and 46-bit virtual address. Assume every page table exactly fits into a single page. If page table entry size is 4B then how many levels of page tables would be required.



$$\text{no. of p.t. entries per page} = \frac{8\text{KB}}{4\text{B}} = 2\text{k} = 2^{11}$$



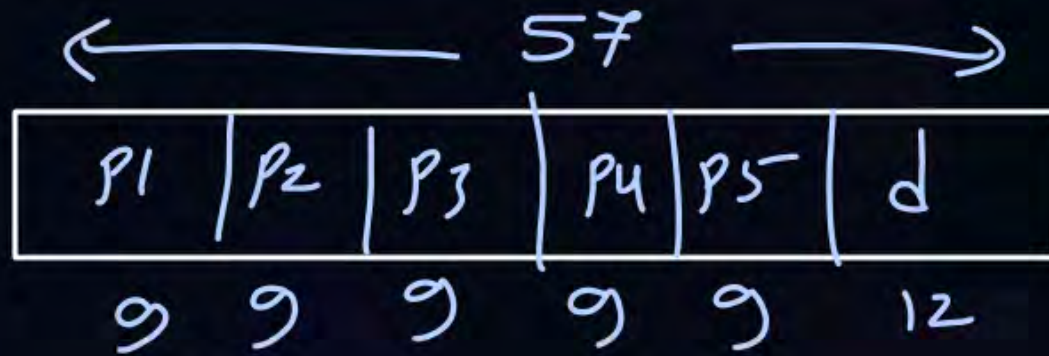
Topic : Question

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- #Q. Consider a computer system with 57-bit virtual addressing using multi-level tree-structured page tables with L levels for virtual to physical address translation. The page size is 4KB ($1\text{KB}=1024\text{B}$) and a page table entry at any of the levels occupies 8 bytes. $\hookrightarrow 2^{12}\text{B}$
The value of L is 5?

$$\text{no. of entries per page} = \frac{4\text{KB}}{8\text{B}} = 2^9$$



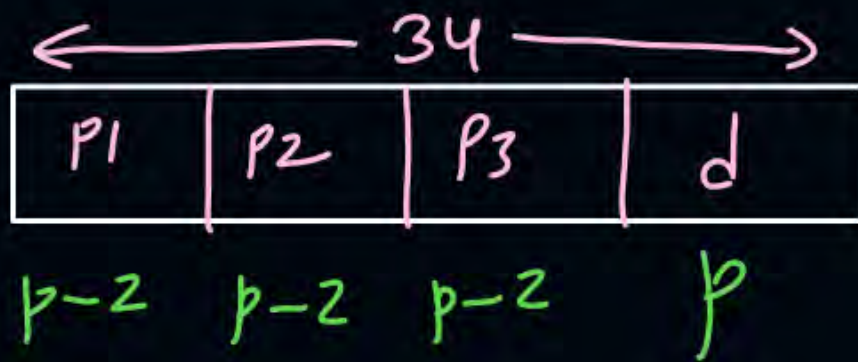
Ques) L.A. = 34 bits

P.T. entry size = 4 bytes

3-levels of paging

outermost page table fits into single page.

page size = 1 kbytes



assume page size = 2^p bytes $\Rightarrow d = p$ bits

$$\text{no. of P.T. entries per page} = \frac{2^p \text{ bytes}}{4 \text{ bytes}} = \frac{2^p}{2^2} = 2^{p-2}$$

$$(p-2) + (p-2) + (p-2) + p = 34 \text{ bits}$$

$$4p - 6 = 34$$

$$4p = 40$$

$$p = 10 \text{ bits}$$

$$\text{page size} = 2^{10} \text{ B} = 1 \text{ KB}$$

Ques) L.A. = 53 bits

P.T. entry size = 8 B

4-levels of paging

outer P.T. fits in exact single page

Page size = 8 kbytes

$$\text{no. of P.T. entries per page} = \frac{2^p}{2^3} = 2^{p-3}$$

53

p_1	p_2	p_3	p_4	d
$p-3$	$p-3$	$p-3$	$p-3$	p

$$4(p-3) + p = 53$$

$$5p = 65$$

$$p = 13 \Rightarrow \text{page size} = 2^{13} \text{ B} = 8 \text{ KB}$$



2 mins Summary

Topic

Frame Allocation

Topic

Thrashing

Topic

Multilevel Paging



Happy Learning

THANK - YOU